# Disambiguating “Population,” “Collective Evolution,” “Convergence,” and “Diversity” in Paper 3

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## Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026), and “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026). It does not introduce new architectural commitments. Its sole contribution is to articulate, in operational form, the precise meanings of four terms — *population*, *collective evolution*, *convergence*, and *diversity* — as those terms appear in Paper 3’s Claim 6, so that downstream work can adopt or argue against the concepts without ambiguity.

## Abstract

Paper 3’s Claim 6 introduces four population-scale terms that do not appear in Papers 1 or 2 as governance concepts: *population*, *collective evolution*, *convergence*, and *diversity*. Each term names a distinct concept and carries a specific scope and register within the Claim 6 projection. This note disambiguates the four terms. The most consequential disambiguation is that *collective evolution* names a projected consequence of accumulated FAI events at population scale — not a governance mechanism and not a designed process. The second key disambiguation is that the balance between *convergence* and *diversity* is not centrally administered; the aggregate balance emerges from individual governance decisions each Self makes about its own participation configuration. The third is that *population* in Paper 3 is defined by architecture adoption, not by enrollment in a designated group. All four concepts carry the calibrated-humility register appropriate to extension claims: they describe what is expected to emerge when many Selves operate under the architecture, not what has been empirically demonstrated. The prior art established by this note covers these four population-scale projections as foreseen consequences of the architecture described across the trilogy.

## 1. Why disambiguation is needed

Paper 3’s Claim 6 closes the trilogy by articulating what the five preceding claims of Paper 3 compose into when their architectural commitments operate at population register. Four terms carry the conceptual work of that closure: *population*, *collective evolution*, *convergence*, and *diversity*. Each term is precise within Claim 6, but each is also a term that appears in adjacent bodies of literature — federated learning, population ecology, complex adaptive systems, organizational theory — where it carries meanings that do not transfer cleanly to the Claim 6 context.

Three confusions are most likely and most consequential. First, *collective evolution* may be read as a governance mechanism — something that is designed, directed, or administered. In Paper 3, it is none of these; it is the expected outcome when the actual intra-Self evolution mechanisms accumulate their effects across many participating Selves over many FAI events. Second, *convergence* and *diversity* may be read as levers that a central authority manages. In Paper 3, neither is centrally managed; both emerge from the aggregate of individual Self governance decisions. Third, *population* may be read as a defined group with membership criteria enforced by a central enrollment authority. In Paper 3, population membership follows from adopting the minimum valid architecture — there is no enrollment body to confer or revoke it.

Clearing these three confusions is the work of this note.

## 2. Population: adoption-defined, not enrollment-defined

In Paper 3 Claim 6, *population* names the aggregate of Selves that have adopted the architecture and participate in FAI events. Two properties characterize the term as it is used there.

First, population membership is determined by having the minimum valid architecture. A Self is part of the population if it instantiates the architectural commitments the trilogy specifies and engages in FAI events under those commitments. No central authority confers membership; no enrollment mechanism maintains a registry. A Self that withdraws from FAI participation exits the functional population without requiring any deregistration process.

Second, population size is indeterminate and variable. Paper 3 does not specify a population size, a minimum viable threshold, or an upper bound. The architectural commitments are designed to support the population-scale dynamics described at Claim 6 whether the number of participating Selves is small or large. Claim 6's projections about what is expected to emerge at population scale are projections about tendencies that would grow more pronounced as adoption expands — not claims that a specific scale has been achieved or is required.

The misreading to avoid is treating *population* as a defined consortium, community, or organization. In Paper 3, the term is a structural description of the aggregate of Selves operating under the architecture — not a social or institutional grouping with governance of its own membership.

## 3. Collective evolution: projected consequence, not governance mechanism

The most important disambiguation in this note is the distinction between *collective evolution* and the intra-Self evolution mechanisms the trilogy specifies.

Papers 1 and 2 establish, within their respective scopes, how individual Selves evolve: through mutation (undirected variation within a Self’s governance structures), directed selection (human-governed updating of knowledge layer content toward preferred patterns), and expression (selection among instinct-layer options for a given execution context). Paper 3’s four-locus evolution-feed mechanism extends these processes by providing a channel through which FAI event outcomes reach each participating Self’s home evolution machinery. These mechanisms — mutation, directed selection, expression, and the FAI-feed locus — are the *actual mechanisms* of evolution in the architecture. They operate within and between individual Selves, under each Self’s governance authority.

*Collective evolution* in Paper 3 Claim 6 names something entirely different: the expected aggregate outcome when these individual-Self mechanisms accumulate their effects across many Selves over many FAI events. When Self A contributes an aspect to a FAI event and Self B absorbs governance patterns from that event into its own directed selection process, Self B’s knowledge structures change in a way that reflects exposure to Self A’s accumulated governance history. When this kind of cross-Self influence compounds across a large population over an extended period, the population’s aggregate governance practices are expected to advance toward broader capability and coordination sophistication. That expectation is what *collective evolution* names.

The critical point is that collective evolution is not governed or designed. There is no population-level governance authority that administers it, no mechanism that targets it, no substrate entry that represents it as a direct object of control. It is a projected emergent property — expected to follow from the architecture operating as specified across many Selves, but not itself a layer of the architecture. The calibrated-humility register is appropriate: this note establishes the projection as prior art, not a demonstrated outcome. The uncertainty is about adoption scale and emergence trajectory; the architectural commitments from which the projection follows are themselves not uncertain.

Confusing collective evolution with a governance mechanism would mischaracterize the architecture in two specific ways: it would imply there is a population-level governance layer that sits above individual Selves and directs evolution across them (there is not), and it would imply that collective evolution can be targeted, activated, or suppressed (it cannot be, in the same way that individual mechanism outputs cannot be).

## 4. Convergence: individual participation governance, aggregate tendency

*Convergence* in Paper 3 Claim 6 names the expected tendency for governance patterns across the population to become more similar as Selves accumulate exposure to each other’s governance knowledge through FAI events and directed-selection absorption.

The structural source of convergence is the directed selection mechanism operating within individual Selves. When a Self absorbs governance patterns from a FAI event into its directed selection process, that Self’s future behavior shifts toward approaches that proved effective in the event’s context. If many Selves independently absorb overlapping patterns from many events, the population’s aggregate governance practices tend toward similarity. That tendency is convergence.

The governance implication is located at the individual Self level, not the population level. Each Self’s home governance authority decides how frequently to participate in FAI events, which aspects to contribute, and which outputs from FAI events to absorb into its own directed selection. These participation configuration decisions together constitute the primary governance lever over how much convergence a given Self experiences. The aggregate convergence rate across the population emerges from the distribution of individual governance decisions across all Selves — no population-level authority manages it directly, and none is specified in the architecture.

This has a practical implication for governance design: a Self that wishes to limit its convergence toward population-wide norms does so through its own participation configuration choices, not through a population-level mechanism that does not exist. Governance of the convergence-diversity balance is therefore always local to each Self, even though its effects are visible at population scope.

### 5. Diversity: home governance sovereignty as structural protection

*Diversity* in Paper 3 Claim 6 names the expected preservation of distinct governance approaches across Selves even as convergence operates. Three architectural features of the trilogy jointly produce this preservation.

First, absorption decisions are home governance’s choice. When a FAI event produces outputs that could update a Self’s knowledge structures, whether and how to absorb those outputs is a decision the Self’s home governance makes. The architecture makes absorption available and governable; it does not require or force it. A Self that declines to absorb FAI outputs retains its prior governance trajectory intact.

Second, the FAI perimeter is additive. Participation in FAI events extends a Self’s coordination capacity at the inter-Self perimeter while leaving the home governance architecture undisturbed. FAI does not replace or override existing governance structures; it adds a coordination layer that dissolves after each event, carrying outputs back to home governance for governed consideration.

Third, mutation stays home. Each Self’s decisions about its own internal evolution — including the undirected variation introduced through mutation, the targets set for directed selection, and the expression patterns selected for instinct-layer execution — are made under home governance authority independent of other Selves’ decisions. There is no mechanism by which population-wide architectural shifts propagate automatically into a Self’s governance structures without home governance authorization.

These three features together mean that even in a large, active population with high convergence tendencies, each Self retains structural sovereignty over its own governance trajectory. Diversity is not a population-level policy decision; it is the expected aggregate property of a population in which home governance sovereignty is preserved at every individual Self. The aggregate balance between convergence and diversity at any moment reflects the distribution of individual governance choices across the population, not the output of any centralized balance-management process.

### 6. Prior-art coverage and calibration

The four terms disambiguated in this note — *population*, *collective evolution*, *convergence*, and *diversity* — are Paper 3 Claim 6 projections about what is expected to emerge when the architecture operates at population scale. They are not mechanisms, not design requirements, and not demonstrated outcomes. They are what the trilogy projects would follow from the architecture if adopted widely and operated over time under the governance commitments the three papers jointly specify.

The prior-art coverage this note establishes is at the projection register. Any system that: defines population membership by architecture adoption rather than enrollment; expects governance pattern similarity to emerge from accumulated cross-Self learning without centrally administering that similarity; preserves governance distinctiveness through home governance sovereignty over absorption decisions; and attributes population-scale governance advancement to aggregated individual-mechanism operation rather than to a designed population-level process — is operating within the territory Paper 3 Claim 6 and this derivation note jointly project. The uncertainty that warrants the calibrated-humility register is about adoption scale and emergence trajectory, not about the architectural commitments that produce these expected dynamics. The commitments themselves are the subject of the source papers; the projections are what follows from them at scale.

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### Source papers


Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration.* April 2026. ORCID: 0009-0004-8065-3235.


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